Smart Mobility Hub — Engineering Report (DRAFT)

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Platform: ROS 2 Jazzy, Ubuntu 24.04, Python rclpy, Docker (verified on Apple Silicon aarch64)

Evidence commit: af37efa (branch integration/current-team-work), 2026-10-21

This is a working draft. Sections for M1-A, M1-B and M5 contain measured results from the repository. Sections marked **TODO** belong to members whose code is not yet in the repository; they must be completed with real results before submission. No result in this report is invented — every number comes from a file in `evidence/` or `src/smart_hub/results/`. Everything is **simulation only**; no physical hardware was used.

1. Requirements and scope

The assignment asks for a simulation-first transport hub in ROS 2: passenger detection (S1), AI demand prediction (S2), bus docking (S3), parking (S4), congestion detection (S5), route replanning (S6), robot escort (S7), integrated coordination (S8) and a negative/fault suite with emergency stop (S9), each verified by tests T01–T18 (§21). The safety rule that shapes the whole design: managers only *publish candidate velocities*; only `safety_supervisor` publishes final motor commands (§4.2).

Scenario	Owner	State at the evidence commit
S1 passenger detection	M1-A	implemented, verified in simulation; real-video evaluation pending
S2 demand prediction	M1-B	implemented, T03 evaluated on held-out seeds
S3 bus docking	M2-A	S3 12/12 live, T05 10/10, T06 passed (code by M5, review by M2-A pending)
S4/S5 parking, congestion	M3	nodes in the ROS package with unit tests; S4 11/11 and S5 9/9 pass live
S6/S7 routing, escort	M4	**TODO — no code yet**
S8 integration	M2-B	coordinator node in the ROS package with unit tests; full run needs S3/S7
S9 faults, e-stop	M5	implemented, verified in simulation (T16, T17, 25-trial repetition)

2. Map, coordinates and frames

Map M1 is a 12 m × 8 m site, origin at the south-west corner, +x east, +y north, metres and radians (§2.1). Queue Q is x ∈ [2, 6], y ∈ [5, 7]; crossing C is x ∈ [4.5, 5.5], y ∈ [1.25, 4.75]; obstacle O is centred at (7, 2) with radius 0.35 m. Graph edge ids are written as sorted node names (`C-J`, `E-J`), a team convention shared by the simulator and the traffic monitor. Frames: `map`, `robot/odom`, `robot/base_link`; map→odom is identity in the simulator. Vehicle ids are `robot_1` and `bus_1`.

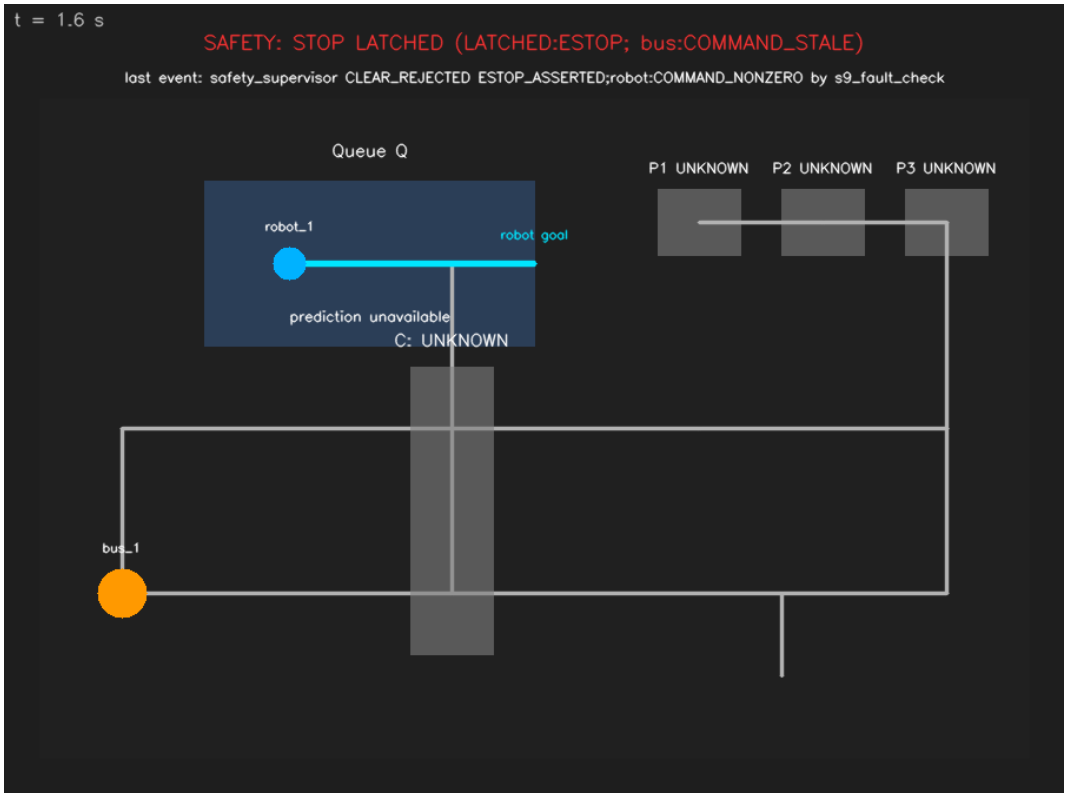


Figure 1 — Dashboard (/hub/markers rendered without RViz) during S9: robot with its route in queue Q, bus at entry A, crossing and parking UNKNOWN because those nodes were not running, safety latched by the e-stop.

3. ROS graph and interface rationale

All custom types are in `smart_hub_interfaces` (14 messages, 6 services, 3 actions, copied field for field from §6–7). The full topic/service inventory with rates, QoS and owners is in `docs/INTEGRATION_CONTRACT.md`.

Interface	Kind	Why this kind
`/hub/queue`, `/hub/demand`, `/hub/vehicles`, `/robot/odom`	topics	continuous state streams with several consumers
`/robot/cmd_vel_candidate` → `/robot/cmd_vel`	topics (R1)	only the newest command matters; the supervisor is the single gate
`/hub/estop`, `/hub/safety`	topics (reliable, transient-local)	a node that starts late must still learn that a stop is active
`/safety/clear_stop`, `/scenario/reset`	services	short request/response with an explicit accept/reject reason
`/bus/dock`, `/robot/escort`	actions	long missions with feedback, cancellation and a result
`/robot/route`, `/bus/route`	topics (`nav_msgs/Path`)	team decision: planned routes for the dashboard and RViz

Every state message is stamped with its *observation* time, and consumers judge freshness by both stamp and receipt time; unstamped or future-stamped data is never fresh (§5).

4. M1-A — passenger detection and queue counting (S1)

Pipeline. `/camera/image_raw` → person boxes → foot point (bottom centre) → planar homography from 4 calibrated floor points (DLT) → nearest-neighbour tracking in metres → count of confirmed tracks inside Q, crossing flag for C → `/hub/queue` at 5 Hz. Two detectors: a documented dark-region baseline (`blob`) for the synthetic camera and OpenCV's pretrained HOG people detector (`hog`) for real video (§8.2 step 1).

Rules (§8.2). 0.4 s appearance and disappearance debounce; identity kept through 0.5 s occlusion; crossing held occupied until C is clear for 2 s; feed silence or a frozen source stamp for 1 s makes the state invalid (count 0, confidence 0, `valid=false`, crossing reported occupied so nobody is granted entry).

Results (simulation, synthetic camera from M5).

Test	Criterion (§8.3)	blob baseline	HOG
T01 stable counts	0, 3, 5, 4 after debounce	0, 3, 5, 4; MAE 0.000 over 127 samples	MAE 0.992 (≤ 1 target met narrowly)
T02 no double count	0.3 s occlusion creates no new id	6 tracks for 6 people	4 tracks for 6 people (recall 0.67)
T02 freeze	invalid within 1.2 s	1.034 s	1.03 s
crossing hold	occupied while in C + 2 s	on after 0.58 s, clear 2.48 s after leaving	same

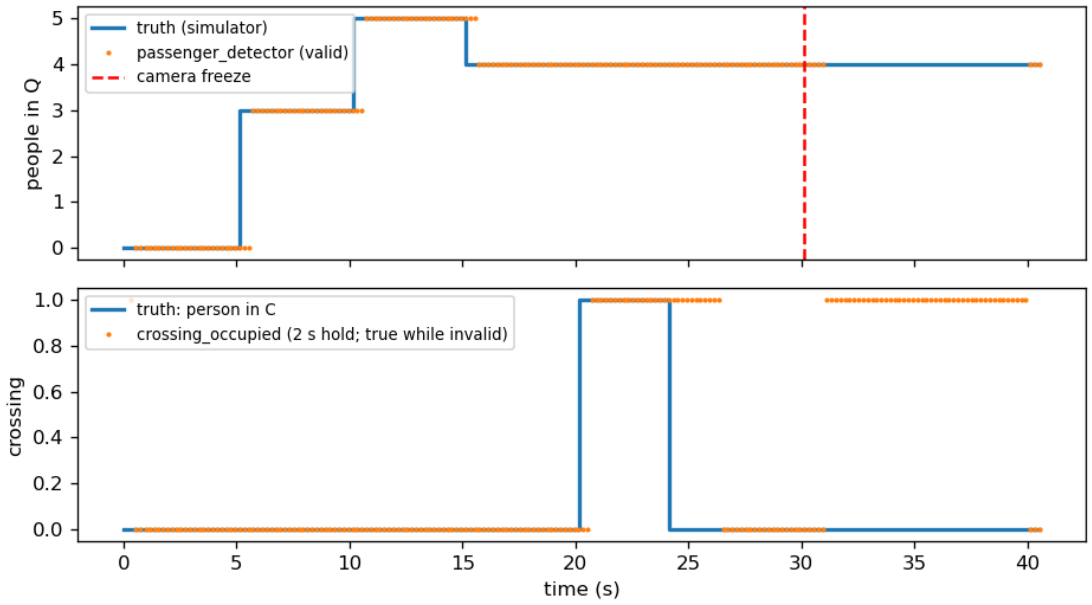


Figure 2 — S1 count-versus-truth (blob baseline). Top: queue count; bottom: crossing flag with the 2 s hold and the conservative "occupied" while invalid after the camera freeze at 30 s.

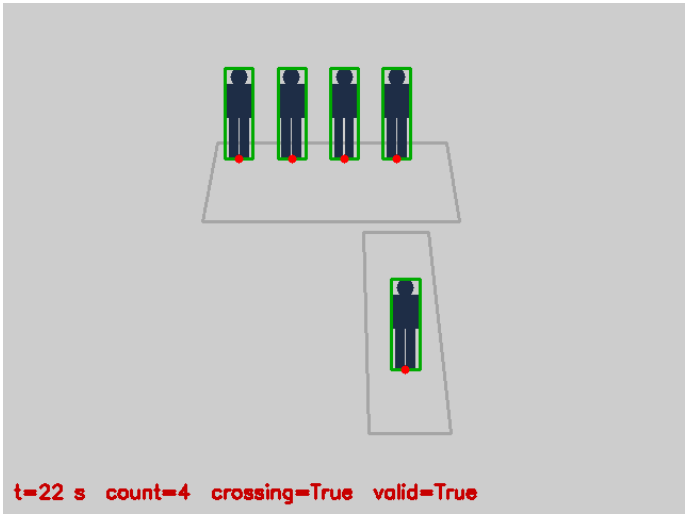


Figure 3 — Annotated synthetic camera frame at t = 22 s: detection boxes, foot points (red), four people in Q and one in C.

A failure found and corrected. The count-versus-truth plot showed that, right after a camera freeze, the count decayed to a *valid* zero for about 0.6 s before the state became invalid, because the disappearance debounce kept running without frames. The count is now judged at the last processed frame; a regression test (`test_silent_feed_never_reports_a_valid_zero`) prevents it from returning. After the fix: 0 valid-zero samples after the freeze.

TODO (M1-A): the §8.3 real-video evaluation — record the lab video, hand-label at least 100 frames, run `video_eval`, and report count MAE, precision/recall and occlusion behaviour here, with annotated before/during/after frames. Tooling is ready and tested (MAE 0.0 on a video rendered from S1).

5. M1-B — AI demand prediction (S2)

Source: `src/smart_hub/results/S2_report.md` and `metrics.json` (synthetic seeded episodes). 50 six-minute episodes over 7 demand profiles, split by whole episode 30/10/10 before windowing; features `q_t`, 10 s and 30 s means, 10 s trend, 30 s arrivals and departures; standardized ridge regression (alpha 30 chosen on validation); persistence baseline. Target: unserved demand at t + 30 s.

T03 (2,500 held-out forecasts)	MAE	RMSE
Ridge, demand target (shipped)	1.215	1.636
Persistence baseline	1.506	2.091

Ridge beats persistence by −0.292 MAE (95 % CI −0.543 … −0.044), better in 7 of 10 test episodes. Inference p95 0.013 ms (target < 100 ms). In a paired closed-loop study over 50 seeds, predictive dispatch lowered mean passenger wait from 73.7 s to 63.7 s (−10.0 s, 95 % CI −14.9 … −6.0). Live in the integrated stack, `/hub/demand` ran at 1.0 Hz on M1-A's `/hub/queue` and became valid after the 60 s warm-up.

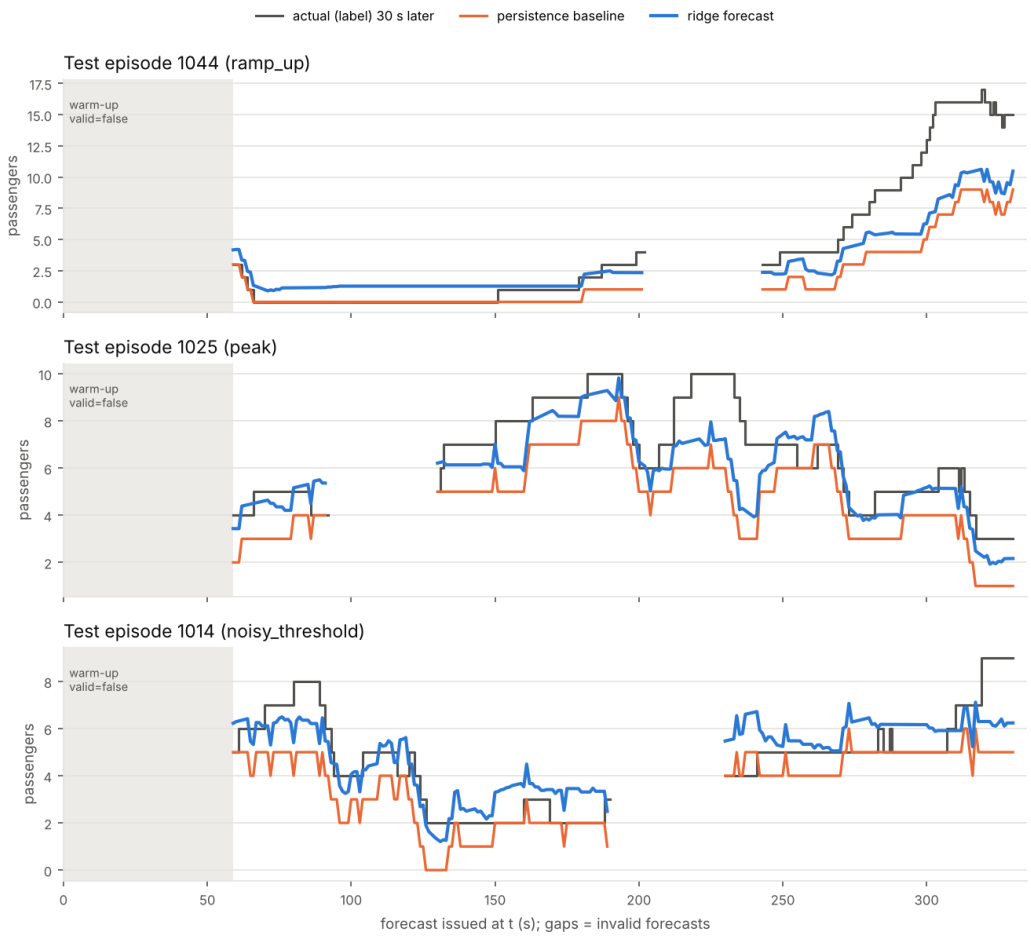


Figure 4 — S2 forecast examples (M1-B).

T04 was run with the coordinator (section 7) using a checker in place of the forecaster; **TODO (M1-B):** restyle files for `ament_flake8`.

6. M2-A — bus manager and docking (S3)

`bus_manager` serves the DockBus action and runs the §10.2 state machine IDLE → APPROACH → (WAIT) → DOCK → BOARD → COMPLETE, with SAFE_STOP from every moving state; the logic lives in `bus_core.py` (pure Python, 10 closed-loop tests against the simulator). The bus follows A-N-Cn-B (fallback while M4's `/route/compute` is missing), asks `/crossing/request` for C 2.5 m before a stop line 0.25 m in front of C, renews every 1 s (3 s lease) and releases once its footprint has left C; when bay B is blocked it waits before C without holding the crossing. Docked = position ≤ 0.20 m, yaw ≤ 0.17 rad and speed < 0.02 m/s for 1 s; the door is requested only then, kept open 8 s from the measured opening, and propulsion needs a measured closed door. ETA = remaining route length / smoothed measured speed + remaining turn time + a declared 1.5 s final-approach allowance; invalid while waiting.

S3 live (`docker/hub.sh s3`, `evidence/m2a/s3/`, 12/12 checks, Figure 5). T05: 10/10 dockings, position error 0.012–0.040 m, yaw error ≤ 0.004 rad, door open 8.1 s per stop with no movement while open, crossing granted before entry and released after exit in all 10, no collision, 4 passengers boarded and 1 left waiting each time. ETA error at 5 m remaining: mean 0.37 s, maximum 1.12 s (target ≤ 3 s); ETA invalid 0.2 s per trip (before a credible speed). T06: zero `/bus/cmd_vel` 0.043 s after the cancel was accepted (limit 0.5 s), status CANCELED, crossing released. A pedestrian in C made the coordinator deny with `PEDESTRIAN_PRESENT`; the bus waited outside C and docked after the person left. A car in bay B gave WAIT with `BAY_BLOCKED` outside C. Freezing the vehicle feed gave a zero candidate after 0.53 s (pose timeout 0.5 s) and SAFE_STOP, then the bus resumed and docked. With seven people in Q the coordinator dispatched the bus by itself (observed-queue fallback) and the mission succeeded. The simulated bus has no inertia, so stopping times are shorter than on hardware.

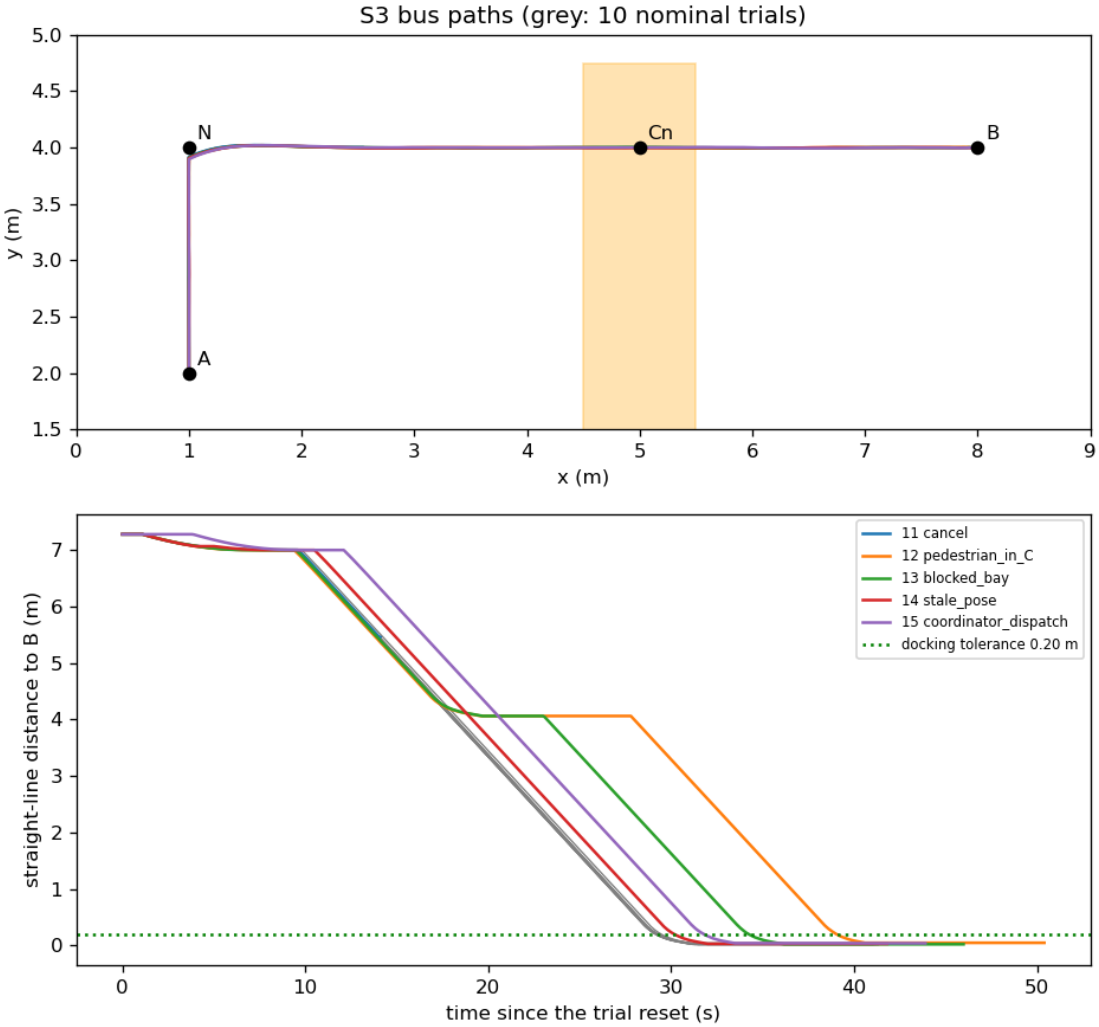


Figure 5 — S3: bus paths (top) and straight-line distance to bay B over time (bottom). Grey: 10 nominal trials; the flat segments at about 4 m are the bus waiting before crossing C (pedestrian in C, blocked bay).

The code and checker were written by M5 for M2-A; **TODO (M2-A):** review them, departure after COMPLETE (a separate transition, §10.2 step 6) is not implemented.

7. M2-B — hub coordinator (S8) — partially TODO

`coordinator_core.py` implements crossing arbitration (one owner, 2 s clear debounce, lease HOLD instead of FREE) and the §9.3 dispatch hysteresis; 28 unit tests pass. Verified against the shared interfaces in a scratch workspace: a crossing was granted while `/hub/safety` was fresh and unlatched, and denied with `SAFETY_STOP` after an e-stop. The node runs inside `hub.launch.py` with the rest of the hub; in S3 it granted and released crossing C for the bus and, with seven people in Q, dispatched the bus by itself.

T04 live (`docker/hub.sh t04`, `evidence/m2b/t04/`, 6/6 passed). A checker publishes `/hub/demand` at 1 Hz and `/hub/queue` at 5 Hz; the real coordinator and bus_manager do the rest. Forecasts alternating around 6 (6.2, 5.8, 6.1, …) for 20 s gave no dispatch. A rising forecast gave exactly one dispatch (reason PREDICTED) 2.0 s after the first value ≥ 6, i.e. on the third consecutive sample; bus_manager accepted one goal and saw no second request while it was outstanding, and none followed in the 10 s after the mission while the forecast was 5 (escalated but below 6). `valid=false` forecasts (predicted 9) and a feed gap with an observed queue of 2 gave no dispatch; with no forecast and an observed queue of 7 the coordinator dispatched once with reason FALLBACK. During the work `demand_freshness_sec` (2.5 s) was added: the forecast arrives at 1 Hz, so with the shared 1.0 s freshness limit a forecast would go stale exactly when the next one was

due (found by review, not observed). **TODO:** the T15 rush-hour run (needs S7).

8. M3 — parking and traffic (S4, S5)

`parking_manager` and `traffic_monitor` with engines and tests (T07–T10): 59 passed, 9 skipped. Against the M5 simulator they produced P1 OCCUPIED, P2 FREE, P3 UNKNOWN (sensor offline) and edge labels from `/hub/vehicles`. Both nodes run inside `hub.launch.py`.

S4 live (`docker/hub.sh s4`, `evidence/m3/s4/s4_check.json`, 11/11 passed). A checker node plays the cars and calls the real services. C1 and C2 asked for P2 at the same moment and received P2 and P3 with different tokens; repeating C1's request id returned the same reservation; a wrong token was rejected with `UNKNOWN_TOKEN` and C2 using C1's token with `VEHICLE_MISMATCH`; C1's slot became OCCUPIED 0.64 s after it was sensed (0.5 s debounce); P3 became UNKNOWN 0.18 s after its sensor went offline; after both reservations expired P2 stayed OCCUPIED and P3 UNKNOWN (expiry never freed an occupied or stale slot); with no free slot a request failed with `FULL`. T07: 20 synchronized request pairs gave 0 double allocations, p95 service latency 39 ms (limit 250 ms), maximum 133 ms.

S5 live (`docker/hub.sh s5`, `evidence/m3/s5/`, 9/9 passed, Figure 6). On edge Cs-J (capacity 4): with one car at 0.35 m/s every label was NORMAL (34 labels, no false congestion); three slow cars joining gave CONGESTED after 0.29 s, held while they stayed; NORMAL 1.8 s after they left. T10: freezing the vehicle feed made every edge UNKNOWN after 0.995 s (limit 1.2 s) with speed and density −1.0, and valid again 0.3 s after the feed resumed. Obstacle O gave BLOCKED after 0.33 s (within one 2 Hz publication) and NORMAL 2.3 s after removal (2 s obstacle-clear hold). The UNKNOWN margin is only 0.2 s because the monitor's stale threshold is 1.0 s.

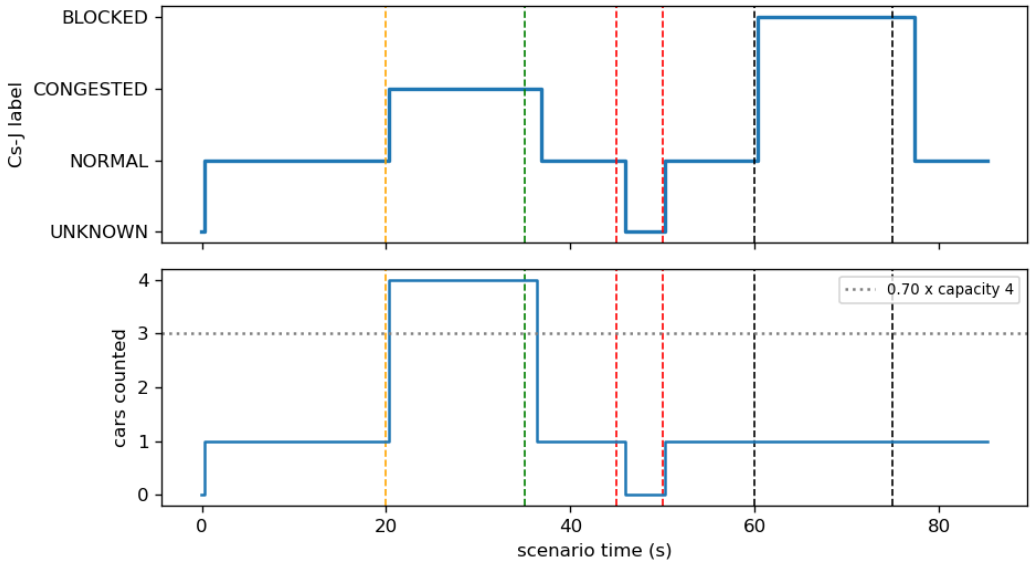


Figure 6 — S5 label of edge Cs-J (top) and cars counted on it (bottom). Dashed lines: slow cars join (orange) and leave (green), vehicle feed frozen and resumed (red), obstacle on and off (black).

The scenario files and both checkers were written by M5 (simulator owner); **TODO (M3):** review them and the results. Observation for the owner: a parked bus at node A is labelled `A-Cs: CONGESTED` because speed 0 counts as slow.

9. M4 — routing and robot escort (S6, S7) — TODO

Route planner (T11 oracle, T12 NO_ROUTE) and robot manager (T13 escort, T14 goal handling). Interfaces ready: `/robot/cmd_vel_candidate` (capped at 0.25 m/s), `/robot/route` drawn on the dashboard, `/crossing/request`. T18 (planner loss) depends on this work.

10. M5 — simulator, safety supervisor and infrastructure (S9)

Components. `scenario_runner` (deterministic 20 Hz unicycle simulator, YAML scenarios with seeded events, synthetic camera, fault injection, `/scenario/reset`); `safety_supervisor` (the only publisher of final commands); `actuator_bridge` (independent motor-side watchdog, caps and door interlock); `metrics_logger` (per-run CSV/JSON); `dashboard_marker_publisher` (read-only `/hub/markers`). Each node is a pure-Python core with unit tests plus a thin ROS wrapper.

Safety policy (§16.2, §18.7). Every 50 ms on a monotonic clock the supervisor evaluates: stop latch, e-stop heartbeat, freshness of candidate (0.5 s), odometry and vehicle data (1.0 s), clearance with a 0.5 s motion look-ahead, site boundary, door interlock and crossing permission; it then clamps to 0.25 m/s (robot) and 0.40 m/s (bus). Two kinds of stop:

Kind	Triggers	Ends
Latch (all agents zero)	e-stop true, footprint outside the site, footprint overlap	only `/safety/clear_stop` when every reset rule passes
Hold (affected agent zero)	stale command/odometry/vehicles/e-stop signal, door, clearance, crossing not granted	automatically when the condition is gone

`/safety/clear_stop` is rejected while the e-stop is asserted or stale, a sensor is stale, a hazard remains, measured speed exceeds 0.02 m/s, or a manager still commands motion; clearing never restarts a mission. A `false` e-stop never clears the latch (§7.1).

Three independent stop layers (T17): supervisor (candidate timeout 0.5 s) → actuator bridge (final-command timeout 0.5 s, own caps) → simulator (drive timeout 0.5 s).

Reset handshake (team decision, §7.1 "after all nodes acknowledge reset"). `/scenario/reset` installs the new world, broadcasts `SCENARIO_RESET_REQUEST` on `/hub/events`, re-sends it every 0.25 s, and answers only when every participant sent `RESET_ACK` (else `MISSING_ACK: <names>` after 2 s). The supervisor keeps its latch across a reset.

Results.

Test	Criterion	Measured
T16 stop latency (§21.3)	e-stop → first zero command ≤ 0.20 s	0.4–1.0 ms over 5 trials (measured outside the supervisor); supervisor receipt → zero ≤ 0.14 ms
simulated speed after e-stop	reported separately (§16.3)	below 0.02 m/s after 0.02–0.10 s
S9 recovery (§16.3)	latch, false keeps latch, reset rejected (e-stop / command / stale odometry), accepted when safe, no auto-restart	13/13 checks passed
T17 watchdog	kill controller, then supervisor	controller killed → supervisor zero after 0.52 s; supervisor killed → bridge zero after 0.50 s; bridge killed → simulator stops
fault repetition (§16.3)	each fault 5× at different mission phases	25/25 passed (table below)
silent continuation	no nonzero output while latched	0 occurrences

Fault (5 phases each)	Passed	Zero-command latency mean / max	Simulated stop max	Limit
e-stop	5/5	0.001 / 0.0012 s	0.10 s	0.20 s
candidate stream lost	5/5	0.513 / 0.534 s	0.60 s	0.75 s
odometry frozen	5/5	1.029 / 1.033 s	1.10 s	1.25 s
vehicle data frozen	5/5	1.032 / 1.041 s	1.10 s	1.25 s
e-stop signal lost	5/5	1.022 / 1.039 s	1.10 s	1.25 s

Phases: start (0.3 s after moving off), cruise 0.20 m/s, turn (0.15 m/s, 0.5 rad/s), maximum speed 0.25 m/s, reverse −0.10 m/s; every trial starts from a fresh world through `/scenario/reset`, so the

table also shows 25 successful reset handshakes. Latencies for the timeout faults equal their design timeouts, as intended; limits add one 50 ms tick plus transport margin.

Failures found and corrected during development. (1) A logger false positive ("motion while latched") came from cross-topic message ordering; the supervisor now publishes its state before its commands and the metric ignores a 0.1 s ordering window. (2) A multi-threaded executor for the reset handshake slowed the simulator from 20 Hz to about 6 Hz and broke S1/S9; replaced by an asynchronous service in its own callback group. (3) The first asynchronous version deadlocked in the default callback group; the unit test now calls the real service. (4) `ros2 launch` exits 0 even when a check fails; `docker/hub.sh s9` now takes its verdict from the evidence file.

Limits. The synthetic camera draws only people; routes appear once M2-A/M4 publish them; hardware mode is refused on purpose (§22 requires an independent power cut); T18 is blocked on M4.

11. Test matrix (T01–T18)

Test	Requirement	Status
T01	S1 counting	PASS (simulation, blob); real video TODO
T02	S1 occlusion / freeze	PASS (simulation)
T03	S2 prediction	PASS (held-out synthetic seeds)
T04	S2 dispatch	pass: live 6/6 (noise, one goal per mission, invalid forecast, feed gap, FALLBACK)
T05, T06	S3 docking, cancel	pass: 10/10 dockings (≤ 0.040 m, ≤ 0.004 rad); cancel → zero 0.043 s
T07–T10	S4, S5	pass: unit tests (M3) and live S4 11/11, S5 9/9 (T07 p95 39 ms, T10 UNKNOWN 0.995 s)
T11–T14	S6, S7	TODO (M4)
T15	S8 integration	TODO (needs all nodes)
T16	S9 stop latch	PASS (simulation)
T17	S9 watchdog	PASS (simulation, process kill)
T18	S9 service loss	BLOCKED (needs M4 planner)

Automated suite at the evidence commit: `colcon test` 187 tests — 185 passed, 2 expected failures that track M1-B's pending style fixes, 0 errors (`evidence/tests/colcon_test_summary.txt`).

12. Evidence and reproducibility

Everything above can be re-run with Docker from a clean checkout: `docker/hub.sh image`, `build`, `test`, `s9`, `s9x`, `s1`. Seeds are fixed (S9 seed 9, repetition seeds 101–125, S1 seed 1); each run records the commit. Files: `evidence/m5/` (S9 check, metrics, rosbag 13.6 s / 1,784 messages, parameter dumps, 25-trial CSV/JSON, dashboard video and frames), `evidence/m1a/` (S1 checks for both detectors, time series, plot, annotated frames), `src/smart_hub/results/` (M1-B).

13. Limitations

All results are from the simulator; no physical robot, camera or barrier was used, and "stopped" means the simulated velocity. Synthetic people are simple figures, which favours the blob baseline and handicaps HOG. Scenarios S6–S8 are not yet implemented. A bus waiting for a pedestrian in C was demonstrated in S3 (detector → coordinator → bus); the robot equivalent needs S7.

14. Contribution table — TODO (each member fills their own row)

Member	Role	Implemented	Tests owned	Reviewed	How the work was done (declare tools, incl. AI assistance, per course policy)
Husan	M1-A	passenger_detector, perception/, video tools	T01, T02	TODO	TODO
Tuymurod	M1-B	demand_predictor, training pipeline	T03	TODO	TODO
Fazliddin	M2-A	TODO	T05, T06	TODO	TODO
Arkar	M2-B	coordinator_core, hub_coordinator	T04, T15	TODO	TODO
Carla	M3	parking_manager, traffic_monitor	T07–T10	TODO	TODO
Aidil	M4	TODO	T11–T14	TODO	TODO
Saidumar	M5	interfaces, simulator, safety supervisor, actuator bridge, metrics, dashboard, Docker, docs	T16, T17, T18	TODO	TODO